

Lecture Notes: Linear Regression

I. What is Linear Regression?

Linear Regression is a fundamental statistical method used for two main purposes:

1. **Prediction:** To predict a continuous quantitative value (like `price`, `weight`, or `size`) based on one or more input variables.
2. **Inference:** To understand the *strength* and *nature* of the relationship between variables (e.g., "How much does `size` *really* change for every 1kg increase in `weight` ?").

The goal is to find the single **"best-fit" straight line** that describes the relationship between an **independent variable (X)** and a **dependent variable (Y)**.

- **Dependent Variable (Y):** The variable we are trying to predict (e.g., Mouse Size). Plotted on the Y-axis.
- **Independent Variable (X):** The variable we are using to make the prediction (e.g., Mouse Weight). Plotted on the X-axis.

II. How Does It Work?

1. The Equation of the Line

The model assumes that the relationship between X and Y can be described by this simple equation:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

- Y_i : The *actual* value of the i -th data point (e.g., the actual size of the 5th mouse).
- β_0 (**beta-naught**): The **intercept**. This is the predicted value of Y when X is 0. It's where the line crosses the Y-axis.
- β_1 (**beta-one**): The **slope** or **coefficient**. This is the most important part. It tells us how much Y is predicted to change for every *one-unit* increase in X .
- X_i : The value of the i -th data point's X variable (e.g., the weight of the 5th mouse).
- ϵ_i (**epsilon**): The **error term** or **residual**. This is the "unexplained" part. It's the difference between the *actual* value (Y_i) and the *predicted* value on the line.

2. Finding the "Best" Line: Ordinary Least Squares (OLS)

How do we find the "best" values for β_0 and β_1 ? We use a method called **Ordinary Least Squares (OLS)**.

The "residual" (ϵ) is the vertical distance from a data point to our line.

- Residual = (Actual Y) - (Predicted Y)

OLS finds the *one* possible line where the **sum of the squares of all the residuals** is at its absolute minimum.

$$\text{Minimize } \sum_{i=1}^n (\epsilon_i)^2 = \sum_{i=1}^n (Y_i - (\beta_0 + \beta_1 X_i))^2$$

This is the exact same "Sum of Squared Residuals" (SS_{residual}) we used to calculate R^2 . Linear regression is the method that *finds* the line that gives you the lowest possible SS_{residual} .

III. The 4 Key Assumptions of Linear Regression

This is the most critical part of linear regression. For your model to be trustworthy and for the p-values and confidence intervals to be valid, **four assumptions about the residuals (ϵ) must be true**.

A common acronym to remember them is **LINE**:

1. L — Linearity

- **What it is:** The relationship between the X variable and the Y variable must be linear.
- **Why it matters:** If the true relationship is a curve (e.g., $Y = X^2$), trying to fit a straight line to it will give you a terrible, useless model.
- **How to check:** Create a scatter plot of X vs. Y. You should see a general straight-line pattern, not a clear U-shape or S-shape.

2. I — Independence

- **What it is:** The residuals (ϵ) must be independent of each other.
- **Why it matters:** This means that the error of one data point does not give you a clue about the error of the next data point.
- **When it fails:** This assumption is most commonly violated in **time-series data**. For example, in stock prices, a high (positive error) day is more likely to be followed by another high day. This "autocorrelation" breaks the independence rule.
- **How to check:** Plot the residuals in the order they were collected. You should see random noise, not a clear "wavy" or "drifting" pattern.

3. N — Normality

- **What it is:** The residuals (ϵ) must be **normally distributed**.
- **This is a common mistake:** The *data itself* (your X's and Y's) does *not* have to be normally distributed. Only the *errors* need to be.
- **Why it matters:** This assumption is required for all the statistical tests (p-values, t-tests, confidence intervals) to be valid. If the errors aren't normal, your p-values are wrong.
- **How to check:** Create a **histogram** or **Q-Q plot** of your residuals. The histogram should look like a bell curve.

4. E — Equal Variance (Homoscedasticity)

- **What it is:** The variance of the residuals must be constant for all values of X. This is called **Homoscedasticity** (homo = "same", scedasticity = "spread").
- **The opposite:** The opposite, and a common problem, is **Heteroscedasticity** (hetero = "different"). This is when the spread of the errors changes as X changes.
 - *Example:* When predicting income (Y) based on years of education (X), your model might be very accurate for low education (errors are small) but very inaccurate for high education (errors are huge). This results in a "fanning out" or "cone" shape in the residual plot.
- **Why it matters:** Heteroscedasticity makes your model's coefficients seem more precise than they really are, leading to misleadingly small p-values.
- **How to check:** Create a **residual plot** (plot the *predicted Y values* on the x-axis and the *residuals* on the y-axis). You should see a random cloud of points with no clear pattern or shape (like a cone or a U).

Summary

If all 4 **LINE** assumptions are met, your model is valid. If they are not, you cannot trust the p-values or confidence intervals, and you may need to transform your data (e.g., take the log) or use a different, more complex model.